

Reproducible AI-Assisted Earthquake Early Warning Research for Western Seismic Regions

Method Development on a California FDSN Pilot with Transfer Path to European Networks

AMGCR Earthquake Research — D-F1 / D-F2

Swiss certificate programme (AMGCR)

29 July 2026

Why Earthquake Early Warning and open methods matter

EEW: alerts seconds to tens of seconds before strong shaking

P-wave detection, magnitude estimation, ground-motion prediction

European risk: Mediterranean belt, Alps, anthropogenic seismicity

Switzerland/EU expect documented, open methodologies

Europe-focused programme; California as methods laboratory

Primary target: European seismic networks

California IRIS/EarthScope pilot: completed validation lab

EarthESND: reference only (88 tests on full checkout)

Pilot science freeze: 2026-07-29

Gaps this workflow addresses

Few packaged EEW workflows for European FDSN data

Missing trails: raw MiniSEED → ML-ready features

Benchmark code decoupled from Western pilot science

Small-N pilots cannot support operational EEW claims

Objectives and completion status

Primary: reproducible, AI-ready workflow for Europe
O1–O6 completed (through 8×18 features + interpretation)
O4: instrument response 0/8 applied — documented
O7–O9 proposed: Europe, ML, real-time

Literature — onsite EEW

STA/LTA and triggers for P-wave onset

Allen & Kanamori (2003); Hoshihara et al. (2008)

Trade-offs: latency, false alarms, magnitude saturation

European focus: metadata, latency budgets, uncertainty

Literature — ObsPy, FDSN, ML, reproducibility

ObsPy (Krischer et al., 2015)

USGS events + EarthScope waveforms in pilot

EarthESND: ESN/DENN; reference code only here

Manifests, JSON configs, phase reports

Dataset — completed California pilot

8 events; M 4.05–4.87; BHZ; 40 Hz; 300 s from origin

CI.ADO (6) · CI.USC (2)

12 MiniSEED on disk at EDA; 8 manifest-linked analyses

No European waveforms in completed pilot

How data were acquired

USGS: California 2024, M 4.0–7.5 → 41 catalog matches

EarthScope: CI, NC, BK; 8 events retained

Distance-ranked station list — not uniform sampling

Public FDSN; no credentials

California pilot workflow

Done: download → EDA → signal → preprocessing → features

Done: interpretation, D-S1/D-S2, D-F1 report

EarthESND src/ not used for pilot results

Proposed: EU FDSN, larger N, ML, SeedLink

Methods — EDA & signal (Phase A)

EDA: ObsPy inspection; four cohort figures

Signal: peak/RMS, 0–5 s noise proxy, peak SNR

STA/LTA: 0.5 s / 10 s; threshold 2.5

Windows start at origin — no pre-event quiet time

Methods — preprocessing & features (Phase B)

Demean, detrend, 5% Hann; band-pass 0.1–15 Hz

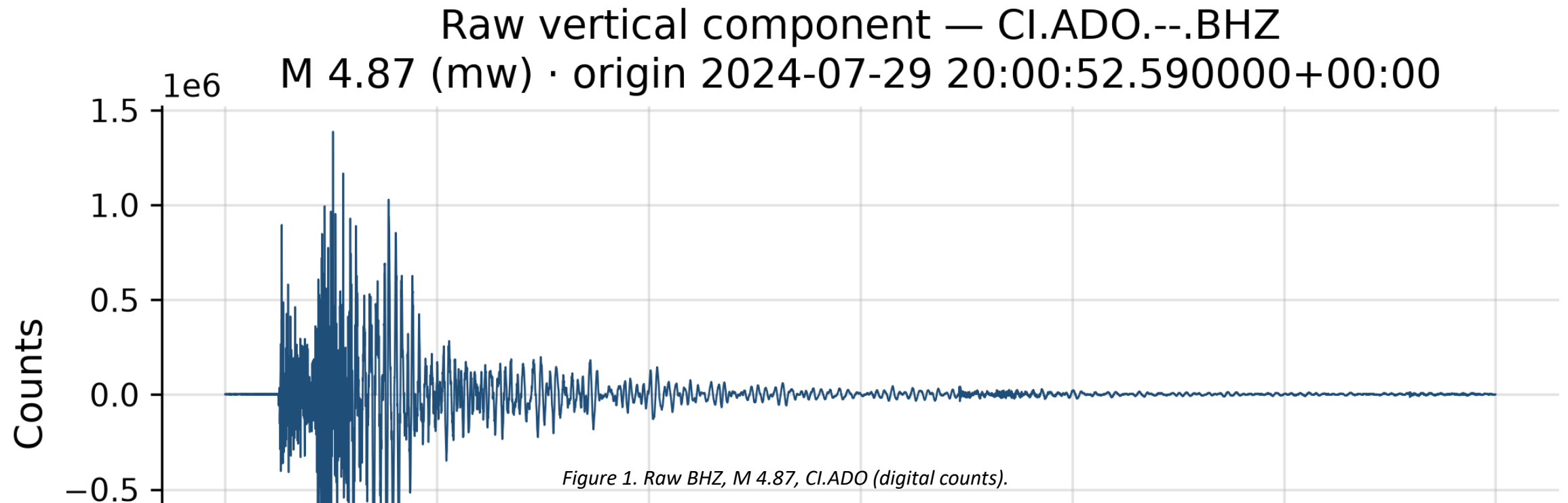
Response removal attempted → 0/8 applied

18 features from filtered traces

PGM/Arias: counts-based proxies only

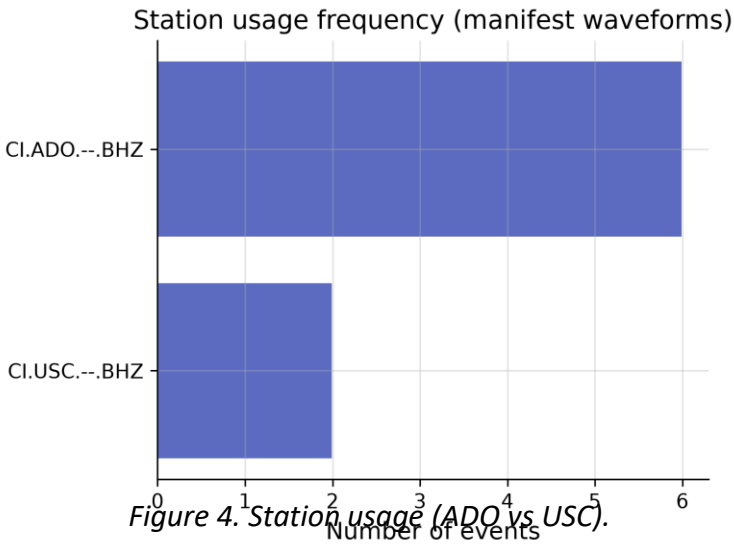
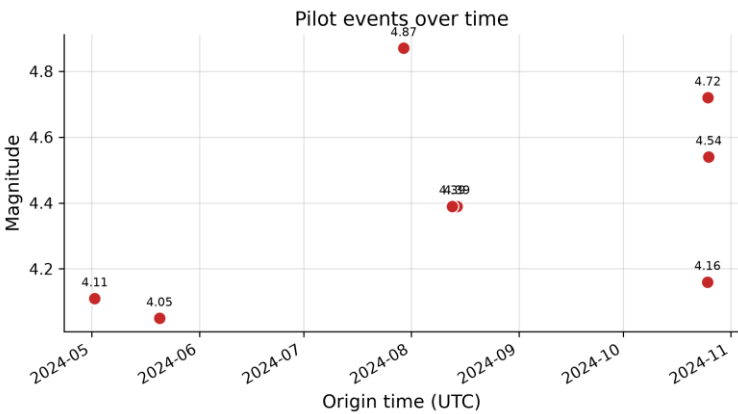
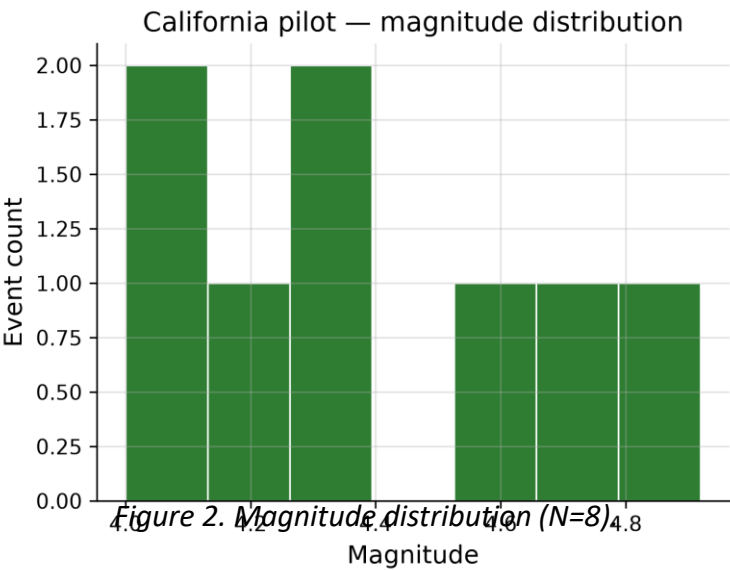
Results — EDA (example waveform)

Figure 1: M 4.87 at CI.ADO — raw counts
Narrow M 4–5 band; query cutoffs
300 s window from trace start



Results — EDA (catalog views)

Figures 2–4: magnitude, time coverage, station usage
ADO supplies 6/8 traces — acquisition bias
Gaps reflect download success

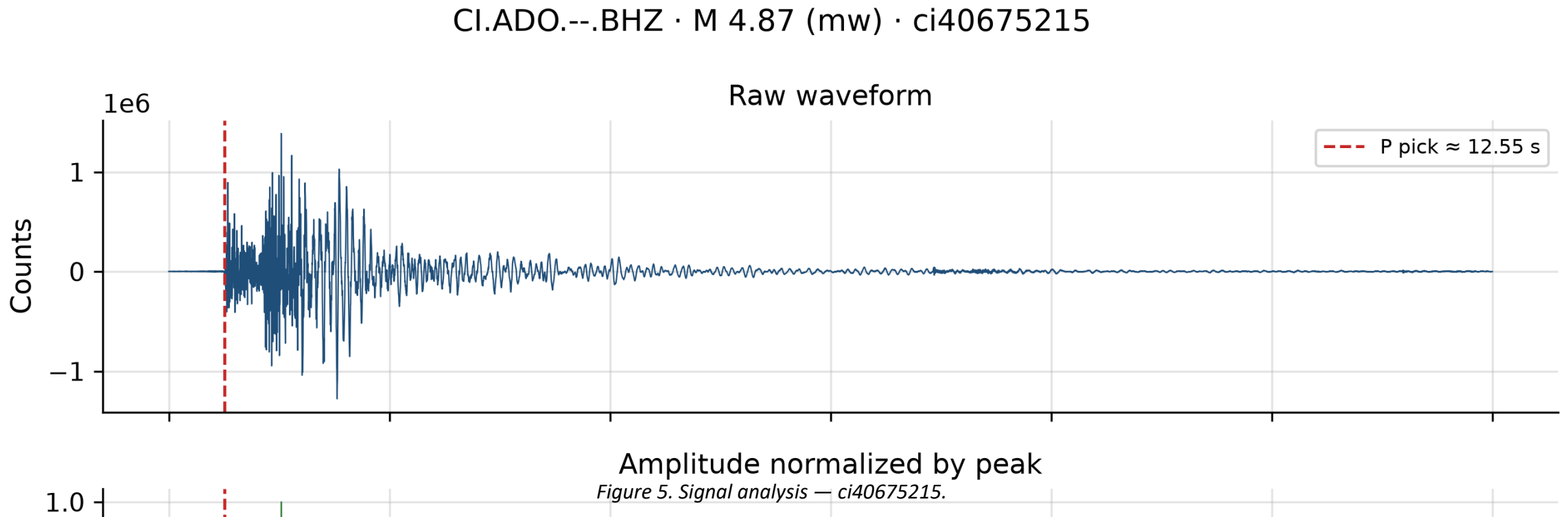


Results — signal analysis (ci40675215)

P pick: 10.15–55.15 s (mean 31.3 s)

Peak SNR: 6.3–778 (median 49.9)

Figure 5: high-SNR case, early pick ~10.2 s

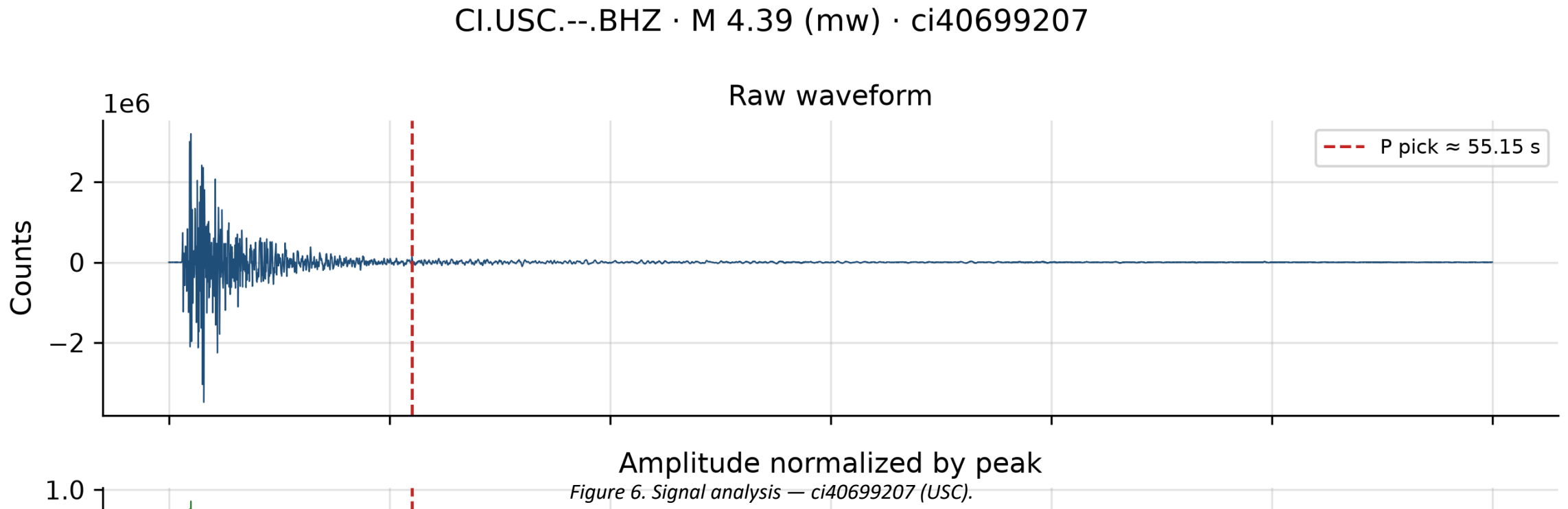


Results — signal analysis (ci40699207 / USC)

ADO: mean SNR 169; USC: mean SNR 42

Figure 6: SNR ~ 6 ; duration 0 s

SNR is QC indicator, not detection probability

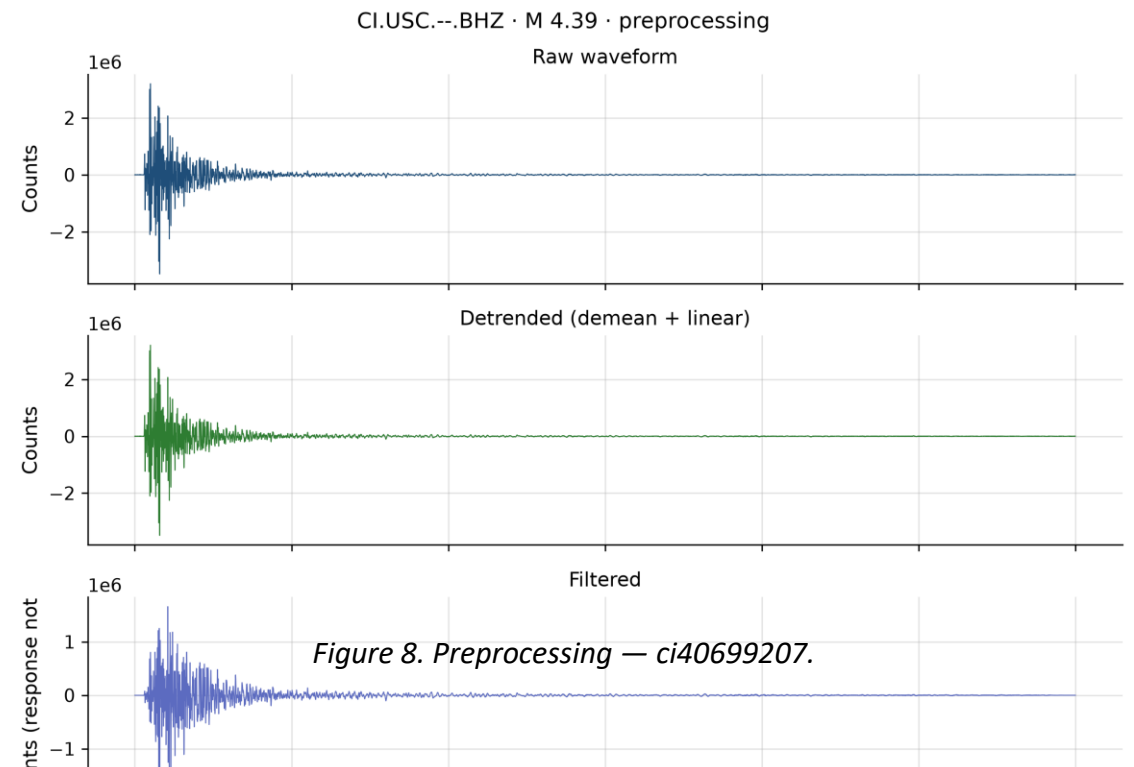
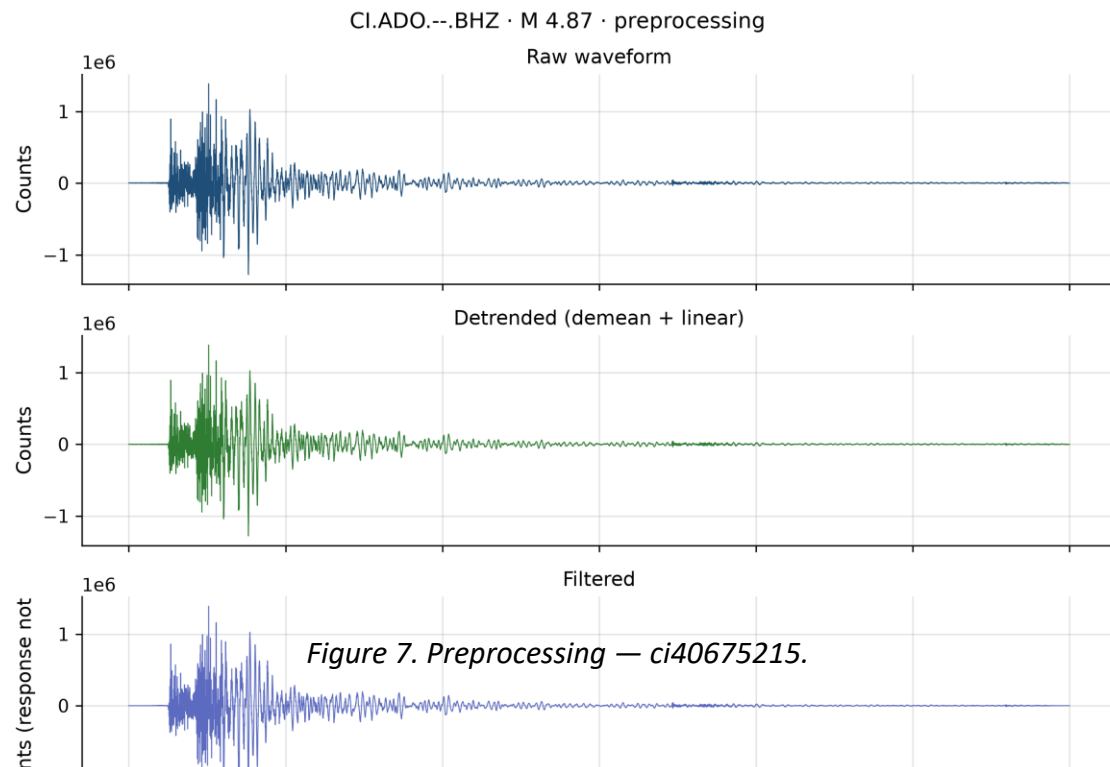


Results — preprocessing

Mean peak-SNR: 137 $\rightarrow$ 1684 (filtered)

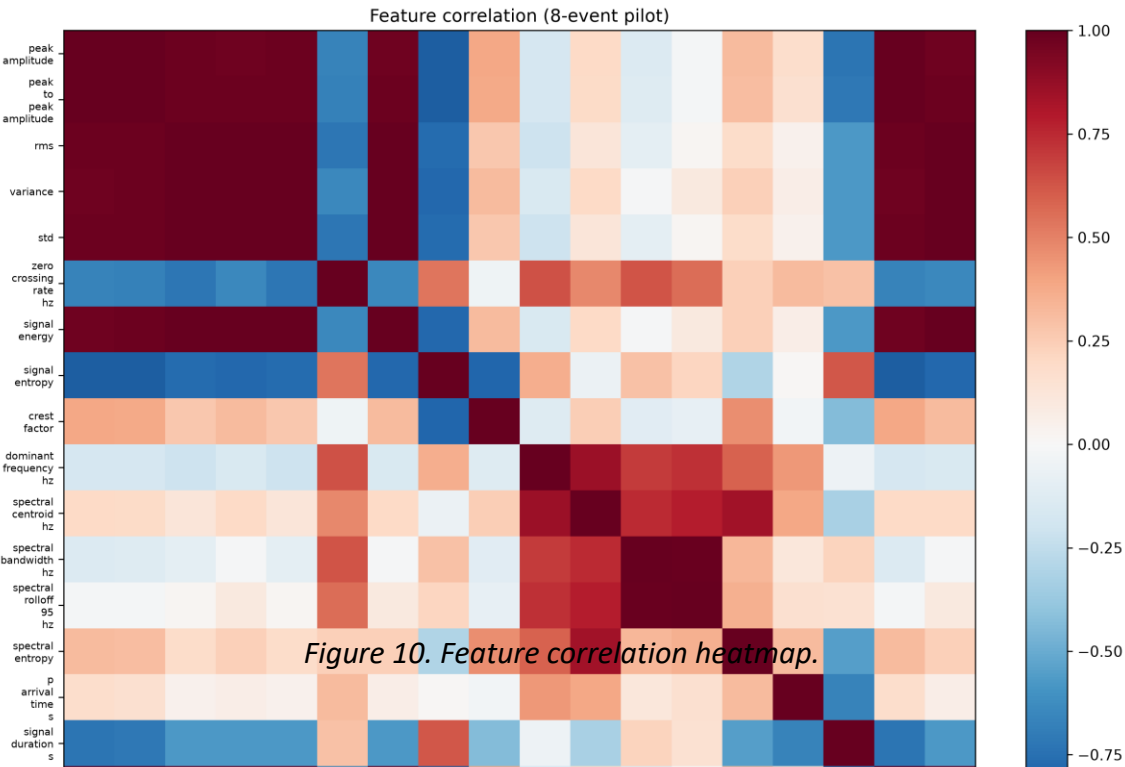
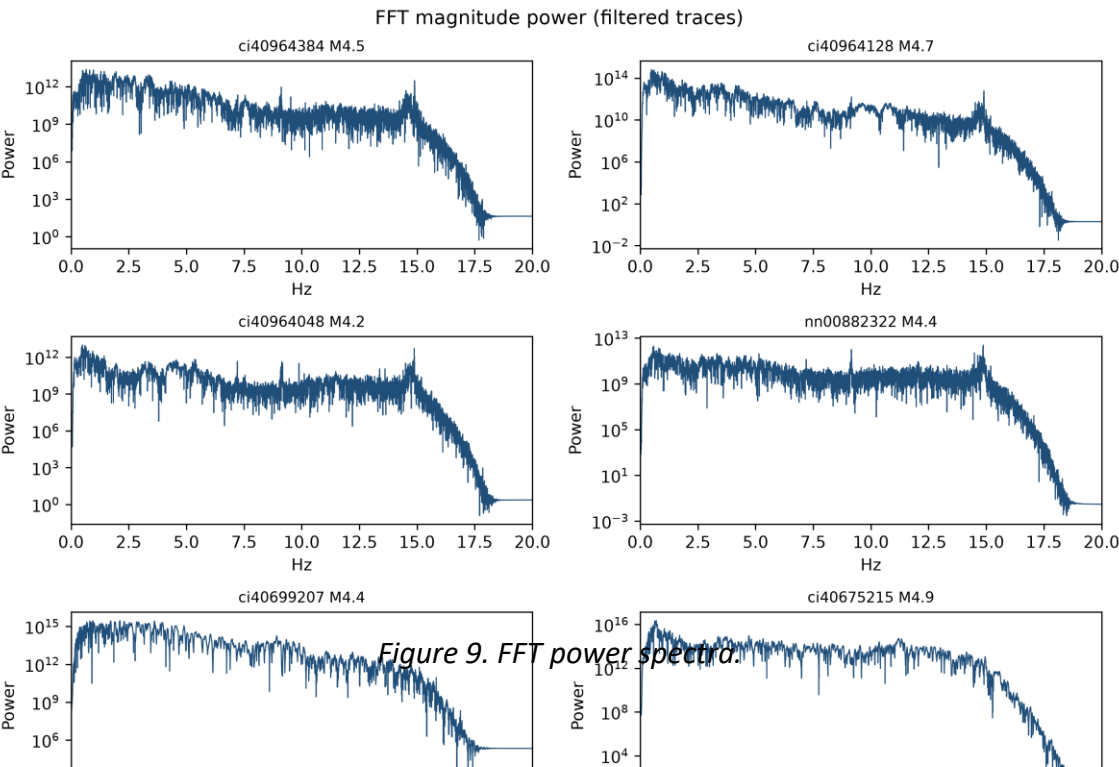
Figures 7–8: raw $\rightarrow$ detrended $\rightarrow$ filtered $\rightarrow$ normalised

Filtered counts $\neq$ physical motion without response



Results — features (spectra & correlations)

8×18 feature matrix
Crest factor mean ≈ 11.4 ; centroid $\sim 2.3\text{--}2.6$ Hz
Figures 9–10: FFT grid, correlation heatmap
No ML training in completed pilot



Results — feature distributions & stations

Figures 11–12: histograms, ADO vs USC boxplots

Amplitude features: high collinearity

USC n=2 — indicative only

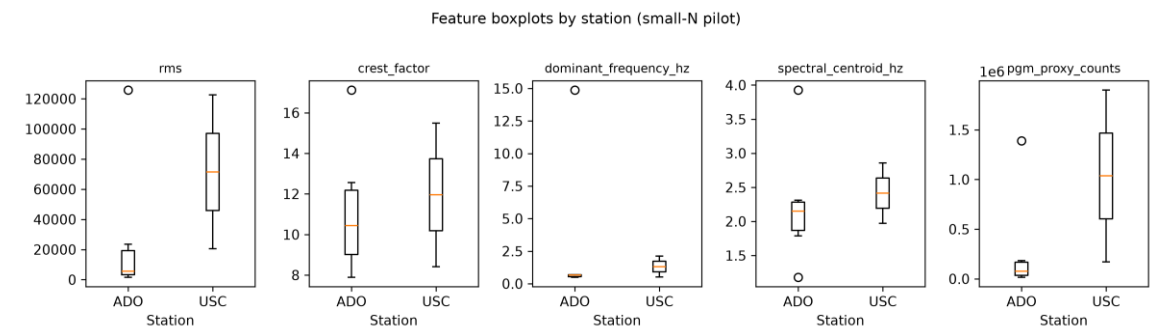
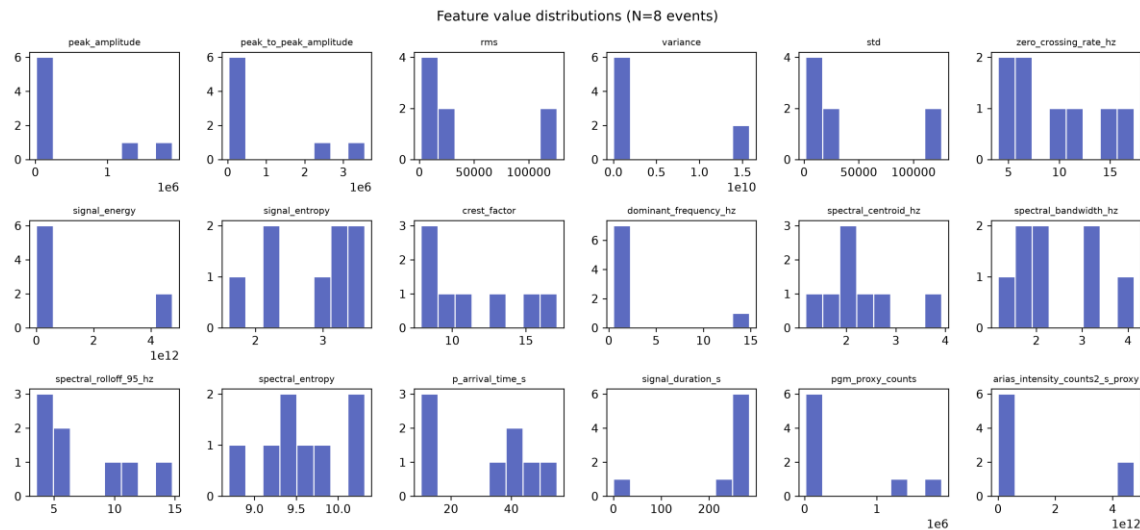


Figure 11. Feature distributions.

Figure 12. Features by station.

Discussion

Script-driven FDSN → features + NPZ stages

Late picks + origin noise limit trigger tuning

Feature matrix: exploratory baselines only

EarthESND: future comparison, not validated here

Limitations

N=8; single BHZ; origin-aligned windows
0/8 response removal; ADO-heavy sampling
Mixed mw/ml; unvalidated P picks
Not comparable to EarthESND K-NET metrics

Threats to validity

Construct: counts vs physical units

External: California $\neq$ Europe $\rightarrow$ EIDA/ORFEUS pilot

Statistical: $N=8 \rightarrow$ larger catalogues

Mitigations are proposed future work

European context & transfer

Europe primary focus; California validates methods

ObsPy/FDSN reproducibility demonstrated

Proposed: ORFEUS/EIDA manifests + same scripts

No EU waveform data yet

Future work, conclusions & thank you

Proposed: 50–200+ events, EU pilot, response fix, ML, SeedLink

Conclusion: reproducible chain to 8×18 features established

Acknowledgements: USGS, EarthScope, ObsPy

Questions?